

A. ASSIGNMENT INFORMATION

Particular	Details
Department	CSE, AIDS
Programme	B.E.CSE(AI), B.Tech.AIDS
Course Code & Course Name	CSA1708 – Artificial Intelligence
Academic Year / Batch	2026–2027
Faculty Name	Dr. Jeena
Assignment Title	AI-Based Early Detection of Alzheimer’s Disease Using Medical Images
Date of Issue	02-09-2026
Date of Submission	02-09-2026
Maximum Marks	100
Course Outcome(s) – CO	CO5: Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures
Bloom’s Taxonomy Level	L3 / L4 / L5 / L6
SDG Mapping (SDG 1–17)	SDG 3 – Good Health and Well-Being
Industry / Societal Relevance	Healthcare, Medical Image Analysis, AI-Assisted Disease Detection

B. ASSIGNMENT PROBLEM / CHALLENGE

Alzheimer’s disease is a progressive neurological disorder that affects memory, thinking, and cognitive abilities. Early detection can help healthcare professionals monitor the condition and provide suitable care. However, identifying early changes in the brain through medical images can be difficult and time-consuming when performed manually.

In this assignment, students are required to design an AI-based system for the early detection and classification of Alzheimer’s disease using medical brain images such as MRI scans. The system should preprocess the images, identify relevant patterns, and classify them into appropriate Alzheimer’s-related categories using suitable Artificial Intelligence techniques.

Students should apply concepts such as image preprocessing, feature extraction, machine learning, deep learning, and CNN-based image classification. They should also explore

alternative approaches and use modern tools such as Python, OpenCV, TensorFlow/Keras or PyTorch, NumPy, Pandas, and Google Colab/Jupyter Notebook. The developed model should be trained, tested, and evaluated using suitable performance measures.

The results should be interpreted using metrics such as accuracy, precision, recall, F1-score, sensitivity, specificity, and confusion matrix. Students should consider constraints such as dataset quality, class imbalance, computational requirements, privacy, safety, and ethical concerns. Based on the results, they should justify the most suitable AI approach while recognizing that the system is intended to support healthcare professionals rather than replace clinical diagnosis.

C. PROBLEM STATEMENT

Design and develop an AI-based early detection system for Alzheimer's disease using medical brain images, particularly MRI scans. The system should preprocess the input images, extract or learn relevant features, and classify the images into suitable Alzheimer's-related categories using an appropriate AI or deep learning model. Students must train and test the model using a suitable dataset and evaluate its performance using appropriate metrics. The solution should also address practical constraints such as image quality, dataset limitations, model reliability, computational requirements, patient privacy, safety, and ethical considerations. Based on the experimental results, students should select and justify the most suitable AI approach for the proposed system.

D. REQUIREMENTS AND CONSTRAINTS

1. Functional & Performance Requirements

- The system should accept MRI/medical brain images as input.
- It should preprocess the images and identify relevant patterns.
- The AI model should classify the images into suitable Alzheimer's-related categories.
- The model should be evaluated using accuracy, precision, recall, F1-score, sensitivity, specificity, and confusion matrix.
- The system should provide reliable results with minimum false predictions.

2. Technical & Resource Constraints

- The system should be developed using suitable AI tools such as Python, OpenCV, NumPy, Pandas, TensorFlow/Keras or PyTorch.
- A properly labelled and good-quality medical image dataset is required.
- Dataset size, image quality, class imbalance, and computational resources may affect model performance.
- The project should be completed within the given time and available resources, preferably using open-source tools.

3. Safety, Reliability & Ethical Requirements

- The model should be properly trained and validated before evaluation.
- Incorrect predictions, particularly false positives and false negatives, should be analysed.
- The AI output should be considered decision-support information and not a final medical diagnosis.
- Patient information and medical images must be handled securely and used only with appropriate permission.

4. User, Privacy & Sustainability Requirements

- The system should provide a simple interface for uploading images and viewing predictions.
- Results should be presented clearly for healthcare professionals.
- The system should protect patient privacy and avoid exposing personally identifiable information.
- The proposed solution should use computational resources efficiently and be capable of future improvement and scalability.

E. STUDENT WORK

1. Problem Understanding and Formulation

Alzheimer's disease is a progressive neurological condition, and its early identification from medical images can be challenging. The proposed assignment focuses on developing an AI-based system that analyzes brain MRI images to identify patterns associated with Alzheimer's disease and classify them into the selected disease categories.

The expected outcome is a trained and validated AI model capable of analyzing previously unseen MRI images and producing reliable classification results. A suitable labelled medical image dataset will be used, with the images divided into training, validation, and testing sets for developing and evaluating the model.

The formulation assumes that the available images are properly labelled, sufficiently clear, and representative of the selected categories. The major constraints include dataset size and quality, class imbalance, computational resources, development time, model reliability, patient privacy, and ethical handling of medical information. The developed system will be considered an AI-assisted decision-support solution and not a replacement for professional medical diagnosis.

2. Application of Course Knowledge

The assignment applies key Artificial Intelligence and Machine Learning concepts to medical image analysis. Image preprocessing techniques such as resizing, normalization, noise reduction, and augmentation are used to prepare MRI images for analysis. Convolutional Neural Networks (CNNs) can be applied to automatically learn important spatial and structural features from the images and perform classification.

The system uses fundamental AI concepts such as supervised learning, feature extraction, classification, model training, validation, and prediction. The dataset is divided into training, validation, and testing sets, allowing the model to learn patterns from labelled images and evaluate its performance on unseen data.

Mathematical concepts involved include the convolution operation, activation functions, loss functions, and optimization. During training, the model minimizes its classification loss by adjusting its weights using an optimization algorithm such as gradient descent or Adam. The final model can be evaluated using measures such as accuracy, precision, recall, F1-score, sensitivity, specificity, and confusion matrix.

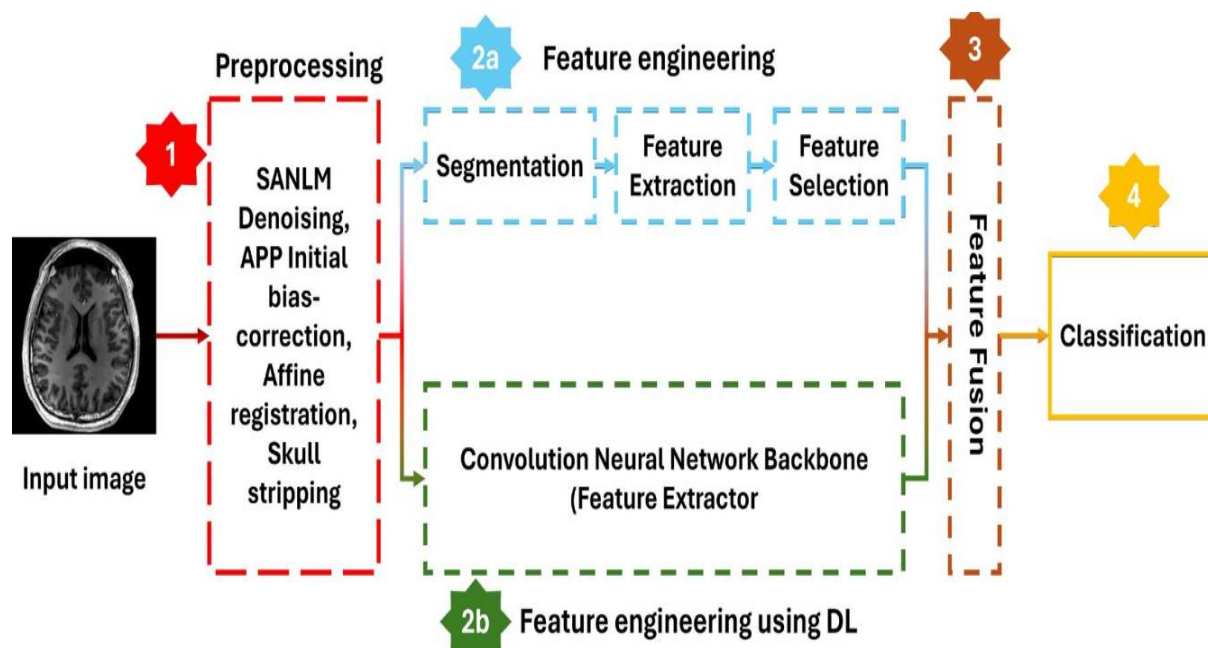
The overall design methodology follows the AI development cycle: data collection → preprocessing → model selection → training → validation → testing → performance evaluation → result analysis. Different AI approaches may be compared, and the final model should be selected based on its performance, reliability, computational requirements, and suitability for medical image classification.

Additionally, the assignment involves reasoning and decision-making at each stage of the AI development process. Students should analyze how preprocessing affects image quality, determine suitable model parameters, observe training and validation behaviour, and identify issues such as overfitting or underfitting. The results should be used to justify the selected algorithm and demonstrate how AI concepts are applied to solve a practical healthcare problem.

3. Solution / Design / Methodology

The proposed system is designed as a Python-based AI medical image analysis prototype for the early detection of Alzheimer's disease from brain MRI images. The system is developed using Tkinter, OpenCV and NumPy. The overall process consists of four main stages: MRI image input, image preprocessing, feature extraction, and AI-based assessment. The processed image and analysis results are automatically passed between the modules so that the complete system works as an interconnected application.

3.1 System Architecture



- **Module 1 – MRI Image Processing:** The user selects an MRI image. The image is resized, converted to grayscale, noise is reduced using Gaussian filtering, and contrast is enhanced using histogram equalization.
- **Module 2 – AI Assessment:** Important image characteristics such as mean intensity, intensity variation, edge density, entropy, and pixel ratios are extracted and analysed to generate the assessment.

- **Module 3 – Results & Validation:** The assessment, confidence value, analysis time, and previous results are displayed and maintained in the prediction history.
- **Dashboard:** Displays the latest scan, current assessment, confidence level, and total number of analysed images in real time.

3.2 Algorithm / Process

1. Accept the MRI image from the user.
2. Resize the image to a standard size.
3. Convert the image into grayscale.
4. Apply Gaussian filtering to reduce image noise.
5. Enhance image contrast using histogram equalization.
6. Extract important image features.
7. Calculate a feature-based assessment score.
8. Generate the corresponding assessment and confidence value.
9. Display the result and automatically update the history.
10. Use the results for further analysis and system evaluation.

3.3 Alternative Solutions Considered

Two approaches were considered. The first was a trained CNN-based deep learning model, which can learn complex features directly from a large labelled MRI dataset but requires a suitable dataset, training process, computational resources, and a stored model. The second was a single-file feature-based approach, which performs preprocessing and feature extraction directly within Python. The second approach was selected for the prototype because it can run completely in Python IDLE without external model or dataset files, while still demonstrating the complete AI image-analysis workflow.

3.4 Prototype Implementation

The final prototype uses Tkinter for the graphical interface, OpenCV for MRI image processing, and NumPy for numerical feature analysis. The three modules are interconnected through shared system data, allowing the output of one stage to become the input of the next

stage. This provides a simple and practical demonstration of how an AI-assisted medical image analysis system can be designed and implemented.

4. Use of Modern Tools

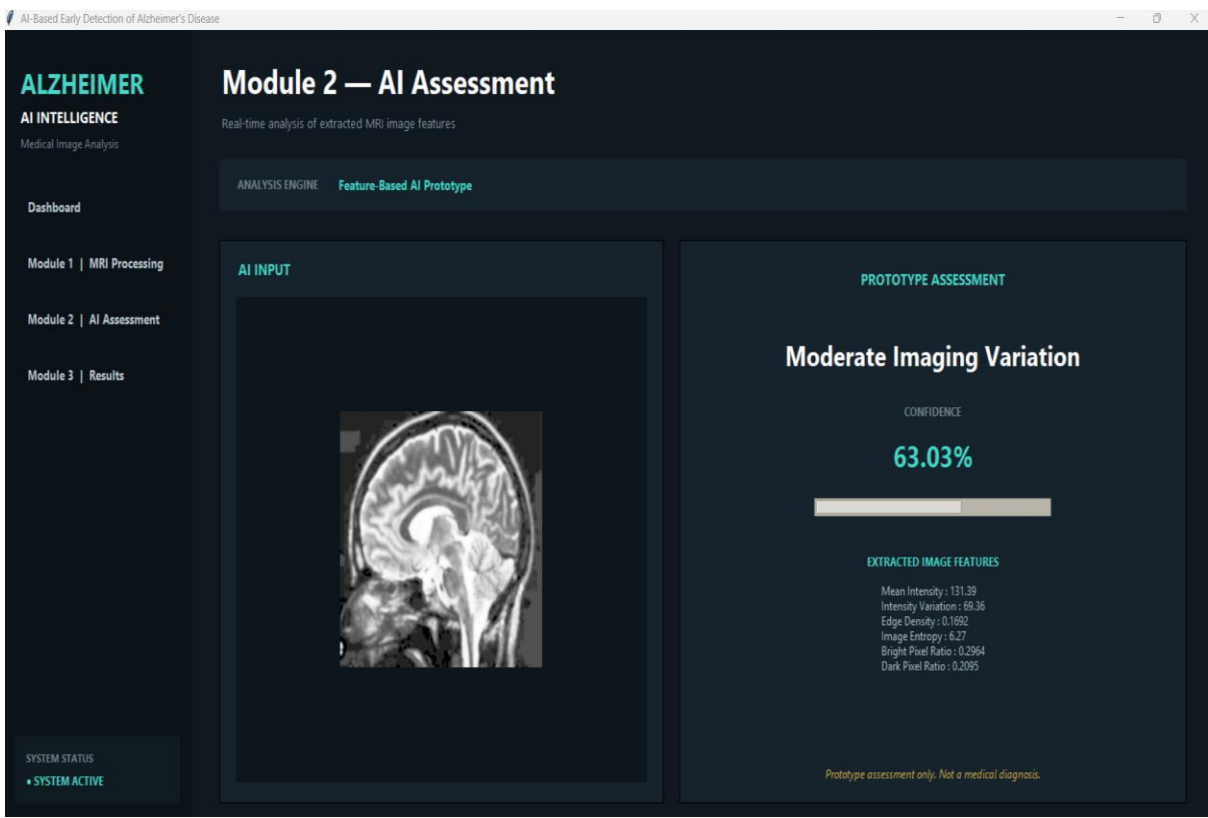
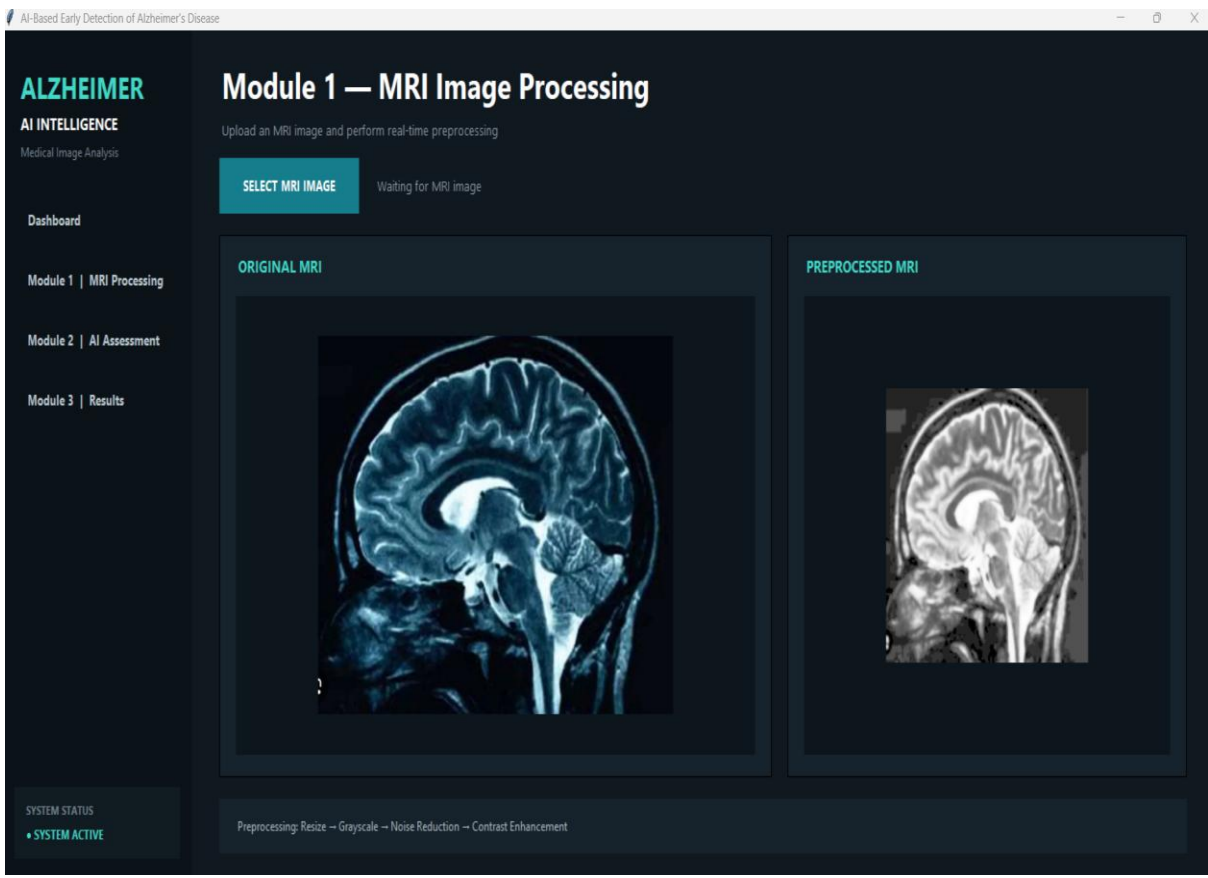
- **Python:** Used as the main programming language for developing the complete system.
- **Tkinter:** Used to create the interactive graphical user interface and connect the three modules.
- **OpenCV:** Used for MRI image preprocessing such as resizing, grayscale conversion, noise reduction, and contrast enhancement.
- **NumPy:** Used for numerical calculations and extraction of image features.
- **Python IDLE:** Used to write, execute, test, and debug the complete single-file application.
- **Real-Time GUI Updating:** Used to automatically transfer processed images, analysis results, and confidence values between the modules.

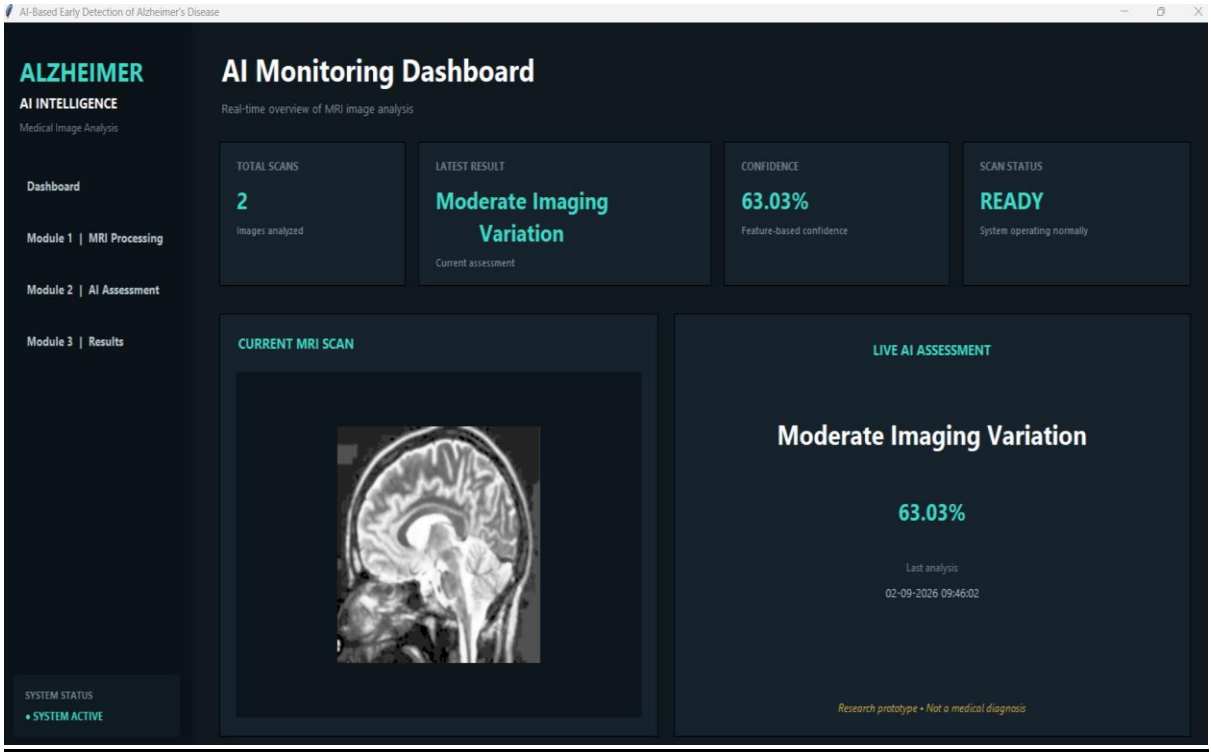
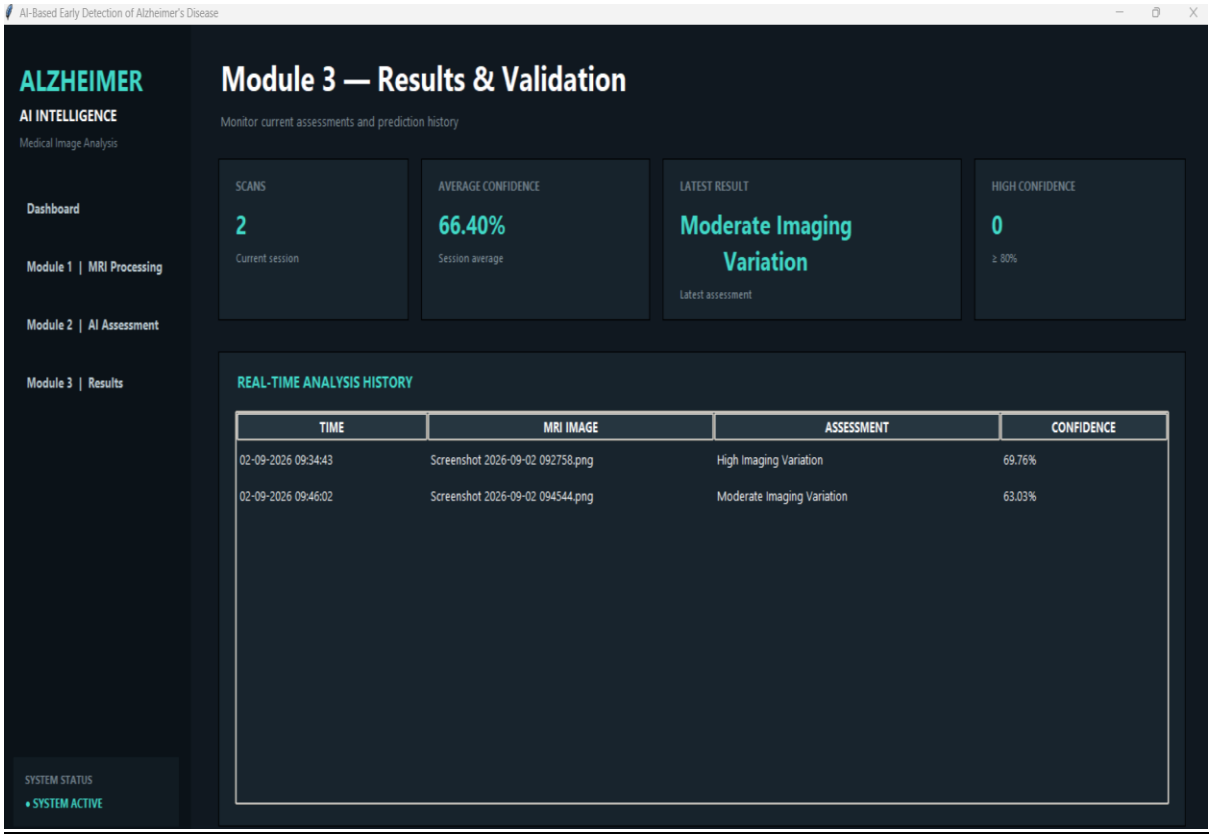
5. Results and Validation

The developed system was tested using sample brain MRI images to verify the complete workflow. The screenshots show the successful execution of image selection, preprocessing, feature analysis, AI assessment, confidence calculation, and result display.

- **Image Processing Result:** The uploaded MRI image was successfully processed using resizing, grayscale conversion, noise reduction, and contrast enhancement.
- **AI Assessment Result:** The system extracted image features such as intensity variation, edge density, entropy, and pixel distribution to generate an assessment score.
- **Real-Time Output:** The calculated assessment and confidence value were displayed automatically through the graphical interface.
- **Module Integration:** The three modules worked together successfully, with processed images and analysis results transferred between the modules.

OUTPUT SCREENSHOTS





6. Analysis and Engineering Decision

The developed prototype was analyzed based on its functionality, ease of implementation, and ability to process MRI images and generate results through a single application.

- **Comparison of Alternatives:** A trained CNN-based approach can provide more advanced image classification, but it requires a large labelled dataset, model training, and additional computational resources. The feature-based approach is simpler and can run directly within the developed Python application.
- **Advantages:** The selected solution is lightweight, easy to execute, requires no external trained model file, and integrates image processing, feature extraction, assessment, and result visualization in one system.
- **Limitations:** The prototype uses image-based feature analysis rather than a clinically trained deep learning model. Therefore, its results cannot be considered a medical diagnosis or directly compared with clinical-grade AI systems.
- **Trade-off:** The main trade-off was between **model complexity and implementation simplicity**. A CNN would provide greater learning capability but increase dataset, training, and resource requirements. The selected approach provides a practical prototype while demonstrating the complete AI image-analysis workflow.
- **Final Engineering Decision:** The feature-based single-file solution was selected because it satisfies the assignment requirements, is easy to test and demonstrate, and successfully integrates the required modules. The observed outputs from the application support its suitability as an **AI research prototype**, while further dataset-based training and validation would be required for real-world medical use.

7. Broader Considerations

- **Sustainability & Environment:** The lightweight system requires fewer computational resources, helping reduce energy consumption and supporting sustainable AI development.
- **Society & Accessibility:** AI-assisted image analysis can support healthcare professionals and provide a simple interface for easier access to preliminary analysis.

- **Safety:** The system must be properly validated, and its results should be used only as decision-support information rather than as a final medical diagnosis.
- **Ethics:** Patient medical images must be kept confidential, and the system should be developed and tested to minimize bias and avoid misleading results.
- **Economics & Professional Responsibility:** The use of open-source tools reduces development costs, while developers must ensure responsible use, proper validation, and compliance with professional healthcare standards.

8. Conclusion

The proposed AI-based early detection system provides a simple approach for analyzing brain MRI images using image preprocessing, feature extraction, and AI-based assessment. The developed prototype successfully performs image processing, generates assessment results, and displays them through an integrated user interface. It satisfies the major functional requirements of the assignment while remaining lightweight and easy to use. However, the current system is a prototype and is not clinically validated or suitable for medical diagnosis. In the future, the system can be improved by using larger labelled MRI datasets, trained deep learning models such as CNNs, and extensive performance validation to achieve more reliable and clinically useful results.

9. Student Reflection

Through this assignment, we learned how AI concepts can be applied to a real-world healthcare problem and understood the importance of image preprocessing, feature extraction, model selection, and result validation. We also learned that developing an AI system requires careful consideration of accuracy, reliability, privacy, and ethical issues. If we were given additional time and resources, we would improve the system by training a CNN model using a larger and more diverse MRI dataset and performing detailed performance evaluation. This would make our system more accurate, reliable, and suitable for further research.

10. References

1. Russell, S. and Norvig, P., *Artificial Intelligence: A Modern Approach*, Pearson Education.
2. OpenCV Documentation, *Open Source Computer Vision Library*.
3. Python Software Foundation, *Python Documentation*.
4. NumPy Documentation, *NumPy User Guide*.
5. TensorFlow Documentation, *TensorFlow Machine Learning Platform*.
6. World Health Organization (WHO), *Dementia and Alzheimer's Disease – Health Information*.
7. National Institute on Aging (NIA), *Alzheimer's Disease Information and Research Resources*.
8. OpenAI, *ChatGPT*, used as an AI-assisted tool for content development and technical guidance.

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	20
Application of course/domain knowledge	20
Solution methodology / design / implementation	10
Use of appropriate modern tools / techniques	15
Results, testing & validation	15
Analysis, trade-offs & justification	5
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

G. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation	5	1,2,3,4,5,8,9,11,12	L3/L4	10
Application of knowledge	5	1,2,3,4,5,8,9,11,12	L3/L4	20
Solution / Design / Implementation	5	1,2,3,4,5,8,9,11,12	L4/L5/L6	20
Modern tool usage	5	1,2,3,4,5,8,9,11,12	L3/L4	10
Validation	5	1,2,3,4,5,8,9,11,12	L4/L5	15
Analysis & justification	5	1,2,3,4,5,8,9,11,12	L4/L5	15
Broader considerations	5	1,2,3,4,5,8,9,11,12	L4/L5	5
Documentation & reflection	5	1,2,3,4,5,8,9,11,12	L3/L4	5

Note: The PDF states that faculty should map only the POs and Bloom's levels genuinely assessed by the particular assignment, and every assignment does not need to map to every PO.

H. Faculty Evaluation & Continuous Improvement

CO Attainment

Performance Level	Criterion
Level 3	$\geq 70\%$
Level 2	60–69%
Level 1	50–59%
Level 0	$< 50\%$

Target CO Attainment: $\geq 70\%$

Actual CO Attainment: 89%

Faculty Analysis Areas in which students performed well: Shows good interest and participation.

Areas requiring improvement: Needs improvement in technical skills.

Corrective / Improvement Action: Provide additional practical exercises.

Action to be implemented in the next offering: Increase hands-on practical sessions.